Question			Answer	Marks	AO	Guidance	
6	(a)		$v = 0$ when $t = 11$ s	B1	3.1b	soi	May be implied by $t = 6$ used in *suvat*
			Distance $= 5 \times 3 + \frac{1}{2} \times 6 \times 3 + \frac{1}{2} \times 4 \times 2$	M1	1.1	Attempt to find the area between the graph and the x-axis Allow ± 2 used for height of the triangle below the axis soi	Trapezium plus triangle gives $\frac{1}{2} \times 3(11 + 5)$ instead of the first two terms
			$[= 15 + 9 + 4] = 28$ [m]	A1	1.1	cao	
			Alternative method for M1A1				
			Distance $=$ $5 \times 3 + \left(6 \times 3 + \frac{1}{2}\left(-\frac{1}{2}\right) \times 6^2\right) + \left\lvert\frac{1}{2}\left(-\frac{1}{2}\right) \times 4^2\right\rvert$	M1		Using *suvat* equation(s) with $a = -\frac{5}{10}$,	Award M1 even if $t = 10$ used
			$[= 15 + 9 + 4] = 28$ [m]	A1			
				[3]			
6	(b)		Car $D - T - 400 = 1400 \times 1.5$	B1	1.1a	No extra forces Allow a used for 1.5 Any form	RHS may be 2100
			Caravan $T - 450 = 900 \times 1.5$	B1	1.1a	No extra forces Allow a used for 1.5 Any form	RHS may be 1350
				[2]			
6	(c)		Caravan $T - 450 = 900 \times 1.5$ gives $T = 1800$	B1	1.1	cao	
			Add the equations $D - 850 = 2300 \times 1.5$	M1	1.1	Solving their equations leading to a value for D soi	May see use of $T = 1800$ directly. Allow for whole system equation soi
			$D = 4300$	A1	1.1	cao	
				[3]			